



BUSINESS
SCHOOL

SKEMA

Corporate Finance PGE M1 Fall 2026

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Session 4: Incremental Cash Flows



About me

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Question



Suppose project *A* requires you to use a machine that you already own, is otherwise unused, and is collecting dust in factory, while project *B* requires you to purchase a new machine. Assume projects *A* and *B* are identical otherwise (cash flows, timing, risk, etc), and the purchase of the machine will not lead to subsequent opportunities. Which project is preferable?



Incremental Cash Flows

For capital budgeting, we care about **incremental cash flows** – the amount by which the firm's free cash flows are expected to change as a result of the investment decision.

Should we care?



You own a facility in Paris. If unused, you could lease out the space for €10m/year in rental income. The new project requires use of this space.

Should we care?



Before starting a new product line, your marketing team launched an investigation into the feasibility of entering this new market. The research cost €2m to complete.

Should we care?



You are a maker of cellphones. Your main product line is the xx Phone 15. You're thinking about expanding into the foldable space. You reckon you'll lose about 10% of sales in the xx Phone series.

Should we care?



There is a small probability that your project, because it is not eco-friendly, will be hit by a lawsuit in the EU and be fined for €500m.

Should we care?



Your project requires the use of a facility. Overhead expenses to keep the facility going run €2m a year. However, you must maintain the facility at all times regardless of whether it is in use.



Earnings vs. Cash Flows

Income statement → *earnings* (“bottom line”)

- ▷ Measure of firm profitability over a period
- ▷ Total earnings of the firm's equity holders (per share: EPS)
- ▷ Keep track of transactions (accountants are historians for firms), use for financial statement analysis (e.g., multiples)

We care about the cash generated by the firm to value it

- ▷ Cash is needed to buy goods, pay employees, fund new investments, pay interests, and dividends to shareholders
- ▷ Cash is not earnings:
 - There are non-cash entries on income statement (depreciation and amortization)
 - Some uses of cash are not reported on income statement (capital expenditures)



Why is there a difference?

Because of the principles governing measurement of *accounting earnings*

Accrual Accounting on income statement...

- ▷ Record revenues/expenses *when a transaction occurs*, **not when they are paid for** (e.g., buying an iPhone with 24 monthly installments)

... implies different treatment of Operating and Capital Expenditures...

- ▷ Operating expenses: only expenses associated with creating revenues in the *current period* → **Δ NWC adjustment**
- ▷ Capital Expenditures: expenses that create benefits over several periods are written off over multiple periods as *depreciation* (fixed assets) or *amortization* (intangibles) → **depreciation and capex adjustment**

From the Income Statement



Sales

- Cost of goods sold (COGS)
- Selling, general, and administrative expenses (SG&A)
- Depreciation

= **EBIT**

- Interest expenses

= **Pretax income**

- Tax

= **Net Income**

Earnings (EPS if divided by number of shares)

Which adjustments to get FCF?

Free Cash Flows: Recipe



Operating Revenue	OPREV
- Operating Expenses	- OPEXP
- Depreciation	- DEPR
= Earnings before interest and taxes	= EBIT
- Tax on Operations	- $\tau \times \text{EBIT}$
= Net Operating Profit After Taxes	= NOPAT
+ Depreciation	+ DEPR
- Capital expenditures	- CAPEX
- Changes in net working capital	- ΔNWC
= Free Cash Flows	= FCF

No interest
expense in FCF!



Working capital (requirement) is the cash consumed in the firm's operating cycle, the unavailable cash tied up in operations

$$\begin{aligned}\text{NWC} &= \text{Operating Cash} + \text{Inventory} + \text{Receivables} + \text{Prepaid} \\ &\quad \text{Expenses} - \text{Payables} - \text{Deferred Revenue} - \text{Accrued Expenses} \\ &= \text{Current Assets} - \text{Current Liabilities} \text{ ("current"=within 1 year)}\end{aligned}$$

*The cash line on the balance sheet represents the total cash balance, so in practice one needs to estimate the operating cash needs of the firm.

Δ NWC



$$\Delta NWC_t = NWC_t - NWC_{t-1}$$

A positive ΔNWC_t means that more cash is tied up in operations, so there is less “free cash”.

Example. Suppose NWC last year was 500. This year, inventory increased by 100 but all other items remained the same. What is the change to free cash flow?

Projecting Free Cash Flows



0 1 2 3 4 5

Revenue

- Costs

- Depreciation

= EBIT

- Taxes

= **NOPAT**

+ Depreciation

- Increase in NWC

- Capex

= **FCF**

PV of FCF

Exercise 1



Home Builder Supply, a retailer in the home improvement industry,... is contemplating whether it should build a new retail store. The company already owns the land for this planned store, which currently has an abandoned warehouse located on it. Last month, the marketing department spent \$10,000 on market research to determine the extent of customer demand for the new store.

Which of the following should be included as part of the incremental earnings for the proposed new retail store?

1. The cost of the land where the store will be located.
2. The cost of demolishing the abandoned warehouse and clearing the lot.

Exercise 1



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Which of the following should be included as part of the incremental earnings for the proposed new retail store?

3. The loss of sales in the existing retail outlet, if customers who previously drove across town to shop at the existing outlet become customers of the new store instead.

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Which of the following should be included as part of the incremental earnings for the proposed new retail store?

4. The \$10,000 in market research spent to evaluate customer demand.
5. Construction costs for the new store.

Exercise 1



Home Builder Supply, a retailer in the home improvement industry,... is contemplating whether it should build a new retail store. The company already owns the land for this planned store, which currently has an abandoned warehouse located on it. Last month, the marketing department spent \$10,000 on market research to determine the extent of customer demand for the new store.

Which of the following should be included as part of the incremental earnings for the proposed new retail store?

6. The value of the land if sold.

Liquidation and Salvage Value



When an asset is sold, any gain on sale is taxed (say, at rate τ). It is recorded in the following way:

Gain on Sale = Sale Price – Book Value

Book Value = Purchase Price – Accumulated Depreciation

Adjustment to FCF = Sale Price – $\tau \times$ Gain on Sale

Example. Suppose you sell land worth €2000 for €4000. Assume $\tau = 40\%$.

Exercise 1



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Which of the following should be included as part of the incremental earnings for the proposed new retail store?

7. Interest expense on the debt borrowed to pay the construction costs.

Exercise 2



You are a manager at Nordex, which is considering expanding its operations in a new generation of wind turbines. Your boss comes into your office, drops a consultant's report on your desk, and complains, "We owe these consultants \$1 million for this report, and I am not sure their analysis makes sense. Before we spend the \$25 million on new equipment needed for this project, look it over and give me your opinion."

Exercise 2



	Project Year				
	1	2	...	9	10
Sales revenue	30,000	30,000		30,000	30,000
– Cost of goods sold	18,000	18,000		18,000	18,000
= Gross profit	12,000	12,000		12,000	12,000
– SG&A	2,000	2,000		2,000	2,000
– Depreciation	2,500	2,500		2,500	2,500
= Net operating income	7,500	7,500		7,500	7,500
– Income tax	2,625	2,625		2,625	2,625
Net Income	4,875	4,875		4,875	4,875

(in thousands of dollars)

Exercise 2



First, you note that the consultants have not factored in the fact that the project will require \$10 million in working capital upfront (year 0), which will be fully recovered in year 10. Next, you see they have attributed \$2 million of selling, general and administrative expenses to the project, but you know that \$1 million of this amount is overhead that will be incurred even if the project is not accepted. Finally, you know that accounting earnings are not the right thing to focus on!

- a. Given the available information, what are the cash flows in years 0 through 10 that should be used to evaluate the proposed project?

Free Cash Flows



0 1 2 .. 9 10

Revenue

- Costs

- Depreciation

= EBIT

- Taxes

= **NOPAT**

+ Depreciation

- Increase in NWC

- Capex

= **FCF**

PV of FCF

Free Cash Flows



0 1 2 .. 9 10

NOPAT

- + Depreciation
- + Adjustment to Overhead
- Increase in NWC
- Capex

= FCF

PV of FCF

Free Cash Flows



	0	1	2	..	9	10
Revenue		30,000	30,000	..	30,000	30,000
- Costs		19,000	19,000	..	19,000	19,000
- Depreciation		2,500	2,500	..	2,500	2,500
= EBIT		8,500	8,500	..	8,500	8,500
- Taxes		2,975	2,975	..	2,975	2,975
= NOPAT		5,525	5,525	..	5,525	5,525
+ Depreciation		2,500	2,500	..	2,500	2,500
- Increase in NWC	10,000			..		-10,000
- Capex	25,000			..		
= FCF	-35,000	8,025	8,025	..	8,025	18,025
PV of FCF						

Free Cash Flows



	0	1	2	..	9	10
NOPAT		4,875	4,875	..	4,875	4,875
+ Depreciation		2,500	2,500	..	2,500	2,500
+ Adjustment to Overhead		650	650	..	650	650
- Increase in NWC	10,000			..		-10,000
- Capex	25,000			..		
= FCF	-35000	8,025	8,025	..	8,025	18,025
PV of FCF						

where

$$\text{Adjustment to Overhead} = (1 - \tau) \times 1000 = 650.$$

Exercise 2



- b. If the cost of capital for this project is 14%, what is your estimate of the value of the new project?
9.56

